Clement **Fung**

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Academic Positions

University of Maryland, Baltimore County — *Baltimore, MD, USA*

Assistant Professor, Computer Science and Electrical Engineering — *08/2026 - present*

Education

Carnegie Mellon University, School of Computer Science — *Pittsburgh, PA, USA*

Ph.D in Societal Computing, Software and Societal Systems Department (GPA: 4.0 / 4.33) — *08/2019 - 09/2025*

- Thesis: "Proposing Guidelines and Approaches to Make Anomaly Detection More Effective for Industrial Control Systems"
  Advisor: Lujo Bauer
- Future Faculty Program, Eberly Center for Teaching Excellence and Educational Innovation
  Completed 2024

University of British Columbia — *Vancouver, BC, Canada*

M.Sc in Computer Science (GPA: 88 / 100) — *09/2016 - 12/2018*

- Thesis: "Dancing in the dark: Private Multi-Party Machine Learning in an Untrusted Setting"
  Advisor: Ivan Beschastnikh

University of Waterloo — *Waterloo, ON, Canada*

B.A.Sc in Systems Design Engineering, Honours (GPA: 88 / 100) — *09/2011 - 05/2016*

- Capstone Project: "Driven: An Automated System for Intelligent Annotation and Analysis of Lane Change Sentiment"
  Advisor: Alexander Wong
  Winner of the GM Canada Innovation Award

Research

Refereed publications

Model Selection of Anomaly Detectors in the Absence of Labeled Validation Data — *FMSD @ ICML 2026*, *Seoul, South Korea*

Alexa Lowe, Clement Fung, Lujo Bauer.
2nd ICML Workshop on Foundation Models for Structured Data.

Model Selection of Anomaly Detectors in the Absence of Labeled Validation Data — *TAI 2025*

Clement Fung, Chen Qiu, Aodong Li, Maja Rudolph.
IEEE Transactions on Artificial Intelligence, Volume 6, Issue 12.

Adopting AI to Protect Industrial Control Systems: Assessing Challenges and Opportunities from the Operators' Perspective — *SOUPS 2025*, *Seattle, WA, USA*

Clement Fung, Eric Zeng, Lujo Bauer.
21st Symposium on Usable Privacy and Security.

Targeted Image Transformation for Improving Robustness in Long Range Aircraft Detection — *IROS 2024*, *Abu Dhabi, UAE*

Rebecca Martin, Clement Fung, Nikhil Varma Keetha, Lujo Bauer, Sebastian Scherer.
2024 IEEE/RSJ International Conference on Intelligent Robots and Systems.

Attributions for ML-based ICS Anomaly Detection: From Theory to Practice — *NDSS 2024*, *San Diego, CA, USA*

Clement Fung, Eric Zeng, Lujo Bauer.
31st Network and Distributed System Security Symposium.

Perspectives from a Comprehensive Evaluation of Reconstruction-based Anomaly Detection in Industrial Control Systems — *ESORICS 2022*, *Copenhagen, Denmark*

Clement Fung, Shreya Srinarasi, Keane Lucas, Hay Bryan Phee, Lujo Bauer.
27th European Symposium on Research in Computer Security.

Biscotti: A Blockchain System for Private and Secure Federated Learning — *TPDS 2021*

Muhammad Shayan, Clement Fung, Chris J.M. Yoon, Ivan Beschastnikh.
IEEE Transactions on Parallel and Distributed Systems, Volume 32, Issue 7.

Towards a Lightweight, Hybrid Approach for Detecting DOM XSS Vulnerabilities with Machine Learning — *WWW 2021*, *Ljubjana, Slovenia (Virtual)*

William Melicher, Clement Fung, Lujo Bauer, Limin Jia.
The Web Conference 2021.

The Limitations of Federated Learning in Sybil Settings — *RAID 2020*, *San Sebastian, Spain (Virtual)*

Clement Fung, Chris J.M Yoon, Ivan Beschastnikh.
23rd International Symposium on Research in Attacks, Intrusions and Defenses.

Brokered Agreements in Multi-Party Machine Learning *APSys 2019*
Clement Fung, Ivan Beschastnikh. *Hangzhou, China*
10th ACM SIGOPS Asia-Pacific Workshop on Systems.

GainForest: Scaling Climate Finance for Forest Conservation using Interpretable Machine Learning on Satellite Imagery *ICML 2019 Workshop*
David Dao, Catherine Cang, Clement Fung, Ming Zhang, Nick Pawlowski, Reuven Gonzales, Nick Beglinger, Ce Zhang *Long Beach, CA, USA*
Climate Change: How Can AI Help?: ICML 2019 Workshop

Other publications

Attacking Autonomous Driving Agents with Adversarial Machine Learning: A Holistic Evaluation with the CARLA Leaderboard *ArXiv 2025*
Henry Wong, Clement Fung, Weiran Lin, Karen Li, Stanley Chen, Lujo Bauer.
ArXiv Preprint: *2511.14876*

Dancing in the Dark: Private Multi-Party Machine Learning in an Untrusted Setting *ArXiv 2018*
Clement Fung, Jamie Koerner, Stewart Grant, Ivan Beschastnikh.
ArXiv Preprint: *1811.09712*

Mitigating Sybils in Federated Learning Poisoning *ArXiv 2018*
Clement Fung, Chris J.M. Yoon, Ivan Beschastnikh.
ArXiv Preprint: *1808.04866*

Invited Talks

Mind the Gap: Improving the Effectiveness of Machine Learning for Industrial Control Systems Security

- George Mason University. April 2026.
- University of Maryland, Baltimore County. April 2026.
- George Washington University. February 2026.
- DePaul University. January 2026.
- New Jersey Institute of Technology. January 2026.
- CyLab Privacy Seminar. January 2026.

Adopting AI to Protect Industrial Control Systems: Assessing Challenges and Opportunities from the Operators' Perspective

- CyLab Partners Conference. October 2025.
- RSTCON 2025. October 2025.
- Symposium on Usable Privacy and Security (SOUPS). August 2025.

Approaches for More Effective ML-based Anomaly Detection in Industrial Control Systems

- Cyber Defense Lab, UMBC. October 2026.
- Secure Cyber-Physical Systems Group, CISPA Helmholtz Center for Information Security. October 2024.
- SysNets Seminar, Max Planck Institute for Software Systems. October 2024.
- Artificial Intelligence and Security Group, Karlsruhe Institute of Technology. October 2024.
- CAE-R Research Symposium, National Cybersecurity Education Colloquium. October 2024.

Attributions for ML-based ICS Anomaly Detection: From Theory to Practice

- RSTCON 2025. October 2025.
- Network and Distributed System Security Symposium (NDSS). February 2024.

Perspectives from a Comprehensive Evaluation of Reconstruction-based Anomaly Detection in Industrial Control Systems

- CAE-R Seminar, April 2023.
- European Symposium on Research in Computer Security (ESORICS). September 2022.

Detecting and Explaining Anomalies in Industrial Control

- Bosch Center for Artificial Intelligence. June 2023.
- Accountable Systems Lab, Carnegie Mellon University. November 2022.
- CyLab Partners Conference. October 2022.

Towards a Lightweight, Hybrid Approach for Detecting DOM XSS Vulnerabilities with Machine Learning

- The Web Conference. April 2021.

The Limitations of Federated Learning in Sybil Settings

- Symposium on Research in Attacks, Intrusions and Defenses (RAID). October 2020.

Brokered Agreements in Multi-Party Machine Learning

- Asia-Pacific Workshop on Systems (APSys). August 2019.

Dancing in the Dark: Private Multi-Party Machine Learning in an Untrusted Setting

- UBC Cybersecurity Summit. May 2018.
- Computer Systems Lab Seminar, University of Toronto. December 2017.

Teaching and Mentoring

Teaching

University of Maryland, Baltimore County

Instructor of Record

- CMSC 491/691: Applied AI for Cybersecurity — Fall 2026

Carnegie Mellon University

Teaching Assistant

- 17-331: Information Security, Privacy, and Policy — Fall 2024
  Instructor: Norman Sadeh
- 11-667: Large Language Models Methods and Applications — Fall 2023
  Instructors: Daphne Ippolito, Chenyan Xiong

University of British Columbia

Teaching Assistant

- DSCI 571: Supervised Learning — Fall 2018
  Instructors: Michael Gelbart, Varada Kolhatkar
- DSCI 523: Data Wrangling — Fall 2018
  Instructors: Jenny Bryan, Rodolfo Lourenzutti
- CPSC 340: Machine Learning — Winter 2018
  Instructor: Michael Gelbart
- CPSC 340: Machine Learning — Fall 2017
  Instructor: Mark Schmidt
- CPSC 210: Software Construction — Winter 2017
  Instructors: Norman Hutchinson, Paul Carter, Mehrdad Oveisi
- CPSC 210: Software Construction — Fall 2016
  Instructors: Norman Hutchinson, Ryan Vogt, Jonatan Schroeder

Mentoring

Carnegie Mellon University

Advised Research Assistants

- Alexa Lowe — Fall ’25, Spring ’26
- Denis Wambold — KIT MS Thesis: visited CMU Fall ’25
- Henry Wong — Summer ’24, Fall ’24, Spring ’25
- Karen Li — Fall ’23, Spring ’24
- Stanley Chen — Spring ’23
- Sneha Rander — Fall ’22
- Jeneel Mashru — Fall ’22
- Kevin Li — Fall ’22
- Shreya Srinarasi — Summer ’21

University of British Columbia

Advised Research Assistants

- Chris J.M. Yoon — Summer ’18, Fall ’18
- Syed Mubashir Iqbal — Summer ’17
- Jaime Koerner — Spring ’17

Professional Experience

Carnegie Mellon University, School of Computer Science — *Pittsburgh, PA, USA*

Postdoctoral Research Associate — *10/2025 - 08/2026*

- Research on AI-based approaches for securing industrial control systems
- Research on the robustness of ML models for aerial drones

Bosch Center for Artificial Intelligence — *Pittsburgh, PA, USA*

Machine Learning Research Intern — *05/2023 - 08/2023*

- Research on applications of diffusion models to improve the performance of visual anomaly detection.
- A paper describing our results was published at IEEE TAI 2025.

Oasis Labs — *Berkeley, CA, USA*

Software Engineer — *01/2019 - 07/2019*

- Developing a secure, private, data-sharing platform for an early-stage blockchain startup.
- A paper describing a use case of this platform was published at an ICML 2019 workshop.

LinkedIn Corporation — *Sunnyvale, CA, USA*

Software Engineering Intern — *06/2015 - 08/2015*

- Search, Network, and Analytics Team: Building infrastructure for online ML-based relevance scoring at scale

LinkedIn Corporation *Mountain View, CA, USA*
Software Engineering Intern *09/2014 - 12/2014*
- Distributed Data Systems Team: Designed and prototyped a derived-data serving system, Venice

Voicebox Technologies *Bellevue, WA, USA*
Software Engineering Intern *01/2014 - 04/2014*
- Server and Tools Team: Developed concurrent database access layer for a mobile voice-assistant service

Ontario Institute for Cancer Research *Toronto, ON, Canada*
Software Developer Intern *05/2013 - 08/2013*
- Developer for the Boutros Lab: a bio-informatics research group

Service

External Service

Committee (Co-)Chair
- USENIX Security Artifact Evaluation Committee, 2027

Program Committee
- RAID 2026
- DSML 2026
- VehicleSec 2026
- ACM SoCC 2026
- CPSS 2026
- ISC 2025
- RAID 2025
- WISW 2024
- ACM FAccT 2023
- ACM FAccT 2022

External Reviewer
- SOUPS 2026
- USENIX Security 2024 (4x)
- IEEE S&P 2024
- IEEE SaTML 2024 (2x)
- USENIX Security 2023 (2x)
- USENIX Security 2022
- IEEE S&P 2021
- NDSS 2021 (2x)
- USENIX Security 2020
- SOUPS 2020
- ISSRE 2017

Invited Reviewer
- IEEE Trans. Dependable and Secure Computing 2025
- IEEE Trans. Information Forensics & Security 2025 (2x)
- Springer Machine Learning 2025
- IEEE Trans. Networking 2024
- ACM CCS Posters 2021
- IEEE Trans. Industrial Informatics 2021

Conference Volunteer
- WOSOC 2024
- SOUPS 2021
- EuroS&P 2021
- EuroS&P 2020

Internal Service

UMBC Department of CSEE
- Graduate Admissions Committee, 2026
- Publicity Committee, 2026

CMU Software and Societal Systems Department
- SCS Graduate Application Support Program, 2025
- Student Representative for Faculty Meetings, 2024–2025
- PhD Student Admissions Committee, 2023
- Faculty Hiring Committee, 2022
- Community Building Committee, 2022–2023
- Prospective PhD Student Visit Day Organizer, 2020–2022

CMU CyLab
- CyLab Sports Committee, 2021–2023

UBC Department of Computer Science
- President of the CS Graduate Student Association, 2017
- Department Strategy Committee, 2017
- Space and Safety Committee, 2017

Awards and Recognitions

2025	**PhD Student Research Award**	RSTCON 2025
2024	**Best Poster Award** *($300)*	IAP CMU Workshop: AI and Security in the Cloud
2024	**AINet Fellow in "Safety and Security of AI"**	Deutscher Akademischer Austauschdienst (DAAD)
2017	**CS Department Graduate Teaching Assistant Award**	University of British Columbia
2017	**CS Department Student Service Award**	University of British Columbia
2016	**Sanford Fleming Award for Co-operative Proficiency**	University of Waterloo
2016	**GM Canada Innovation Award** *($500)*	University of Waterloo
2015	**W.W. King Exchange Fellowship** *($500)*	University of Waterloo
2014	**President's International Experience Award** *($1500)*	University of Waterloo
2013	**Sanford Fleming Award for Outstanding Work Term Report** *($300)*	University of Waterloo
2011	**Colonel Hugh Heasley Engineering Scholarship** *($10000)*	University of Waterloo
2011	**President's Scholarship of Distinction** *($2000)*	University of Waterloo